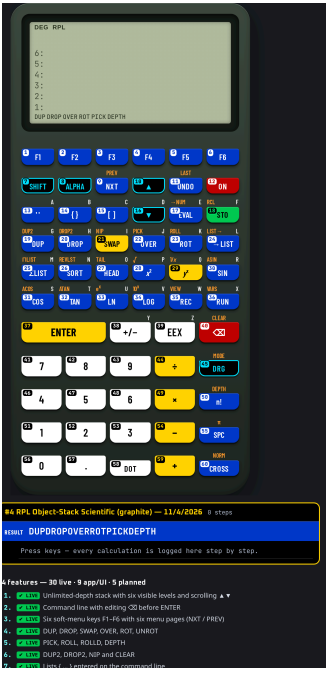
U-45 CALCULATORS

User Manual

Calculator #4 RPL Object-Stack Scientific (graphite)

Scientific – RPN



Complete kit: functions · numbered key map · worked examples · tricks & hacks · shortcuts · common mistakes · practice drills

About this manual

This manual belongs to the **U-45 CALCULATORS** app and describes calculator **#4 — RPL Object-Stack Scientific (graphite)** exactly as it behaves in the app. Every worked example and every drill answer in this book was run on the app's own keys by an automated checker before printing; only results that matched were kept.

How key presses are written

Keys are written as a chip showing the **key number** and its face, for example 

. Numbers follow the numbering printed on the keys (switch it on with "# On"). A shifted function is written as two chips: the shift key, then the key that carries the function.

Entry style of this model

RPL: objects go on an unlimited stack with ENTER; commands act on level 1, 2, …; algebraics in quotes.

Chapters

1. Getting started
2. Features
3. Feature guide (2b)
4. Numbered key map
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6. Worked examples
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1 · Getting started

Open U-45 CALCULATORS, choose **#4** from the home screen (or the jump list at the top), and the calculator appears with its numbered, colour-coded keys. Click or tap keys, or type on a keyboard.

Numbers on every key

Every key carries a small number in the same reading order as this manual (left → right, top → bottom; the navigation pad counts ▲ ▼ ◀ ▶ then its centre). Use "# On / # Off" in the bar above the calculator to show or hide them — the setting is remembered.

High-contrast colours

Keys are coloured by job: white = digits, yellow = operators/execute, blue = functions, green = memory/variables, red = clear/delete/power, black with cyan = shift/modes/menus/navigation. Press "Classic" to return to the original model colours and "Hi-Contrast" to come back.

Three typefaces

Display text uses the U45 Display face, key legends use U45 Keys (a condensed face that stays readable on small keys) and key numbers / helper text use U45 Mono. All three are bundled inside the app, so they work offline.

Copy the display

The Copy button copies exactly what the display shows (indicators that are switched off are skipped) so you can paste a result into a message or spreadsheet.

Fit or 1:1

Fit scales the calculator to your screen; 1:1 shows it at its design size. On a phone, Fit is the comfortable choice.

Reset

Reset rebuilds this calculator from scratch: display, memory, stack and modes return to their start-up state. Use it before each worked example in this manual.

Keyboard

On a computer you can type: digits, + − * / ^ ( ), Enter or = to execute, Backspace to delete, Esc to clear, arrows to move. Letters reach common functions (see the Keyboard shortcuts chapter).

Tooltips

Hover over (or long-press) a key to see "#number name".

Tools rail (v1.4)

A cyan icon rail on the right of every calculator opens 15 panels: Currency (166 currencies, rates auto-updated from open.er-api.com or Frankfurter when online and cached with a date; the bundled dated snapshot is used offline), Units (exact SI factors), Health (published formulas, not medical advice), Simple, Complex, Loans, Construction, Mechanical and Engineering (30+ numbered tools each), Calendar generator, Customize (26 options, saved per calculator), Settings (31 options), History, Hide and View. On narrow phones the rail shows icons only.

History (v1.4)

Every calculation is logged per calculator with a date and time. Open History on the rail to search, re-use (types the input back into the calculator), copy, delete one, select several and delete, or delete all (asks for confirmation). History is stored encrypted on this device (AES-GCM); exporting it needs the passcode.

Camera Solve (v1.4)

The green CS button between Restore and Social reads a problem from the camera, a photo, a PDF or a text file. Recognition (Tesseract OCR, English) runs offline on the device; an online language download is only used if you switch it on in Settings. Check and edit the recognised text, then Solve shows the steps offline. OCR is not available when the app is opened as a plain file:// page in a browser.

Layouts and view modes (v1.4)

The layout selector between Fit and # offers Auto plus 12 layouts: 3D classic, 3D high-contrast, Classic, Pocket, Compact, Expanded, Desktop wide, Split, One-hand left, One-hand right, Large print and Minimal. Auto picks one for your screen and your choice is remembered per calculator. The View panel hides the top bars, the tool bar, the rail or everything (full screen); the round yellow ⤢ button at bottom-left (or Esc) always restores the normal view.

Hide items (v1.4)

The Hide panel lists every button, rail icon and section of this calculator as tick boxes. Hiding is instant; unhiding asks for the app passcode.

Image protection (v1.4)

Right-click, drag and long-press saving of images is blocked inside the app. Saving or sharing an image export (PNG, JPG, QR, calendar), printing a calendar, and downloading or sharing the app itself ask for the passcode, and exported images carry a U-45 watermark. Honest note: this cannot stop a screenshot or screen recording made by the operating system or another device.

Works offline

U-45 CALCULATORS is an offline app: after the first load everything — including fonts — comes from the device.

Colour groups on this calculator

Colour	Group	Keys on this model
num	Digits & number entry	#38, #39, #41, #42, #43, #46, #47, #48, #51, #52, #53, #56, #57, #58
op	Operators & execute	#21, #29, #37, #44, #49, #54, #59
fn	Functions	#1, #2, #3, #4, #5, #6, #9, #11, #13, #14, #15, #17, #19, #20, #22, #23, #24, #25, #26, #27, #28, #30, #31, #32, #33, #34, #35, #36, #50, #55, #60
mem	Memory, variables & registers	#18

clr	**Clear, delete & power**	**#12, #40**
mod	**Shift, modes, menus & navigation**	**#7, #8, #10, #16, #45**

2 · Features

44 features: 30 live on the keys, 9 app/UI features, 5 planned (documented but not yet working — they are clearly marked and never used in examples).

1. LIVE **Unlimited-depth stack with six visible levels and scrolling ▲ ▼**
2. LIVE **Command line with editing ⌫ before ENTER**
3. LIVE **Six soft-menu keys F1–F6 with six menu pages (NXT / PREV)**
4. LIVE **DUP, DROP, SWAP, OVER, ROT, UNROT**
5. LIVE **PICK, ROLL, ROLLD, DEPTH**
6. LIVE **DUP2, DROP2, NIP and CLEAR**
7. LIVE **Lists { … } entered on the command line**
8. LIVE **→LIST and LIST→ build and explode lists**
9. LIVE **ΣLIST and ΠLIST**
10. LIVE **SORT and REVLIST**
11. LIVE **SIZE, GET, HEAD, TAIL**
12. LIVE **ΔLIST first differences and SEQ 1…n**
13. LIVE **Element-wise list arithmetic with + − × ÷ and scalars**
14. LIVE **Vectors […] with DOT, CROSS and NORM**
15. LIVE **MEAN, MAX and MIN of a list**
16. LIVE **Symbolic algebraics in quotes, e.g. '2+3*4'**
17. LIVE **Arithmetic on algebraics builds a new algebraic**
18. LIVE **EVAL / →NUM evaluates an algebraic with stored variables**
19. LIVE **Named variables: value, 'NAME', STO; RCL and typing the name recalls it**
20. LIVE **Registers R0–R9 via STO/RCL + digit**

21. LIVE VARS list and PURGE (one or all)
22. LIVE Keystroke program recorder REC with RUN and step VIEW
23. LIVE UNDO of the last stack operation
24. LIVE LASTARG (LAST X)
25. LIVE Alpha keyboard for names (ALPHA layer)
26. LIVE Scientific functions sin cos tan, ln, log, e^x, 10^x, $\sqrt{}$, x^2, 1/x, n!
27. LIVE Degree / radian / grad modes
28. LIVE Clear error texts: Too Few Arguments, Bad Argument Type
29. LIVE Numbers in standard, fixed, scientific and engineering formats
30. LIVE Matrix entry [[a b][c d]] on the command line
31. APP/UI Every key numbered 1…N in reading order — matches the key map in the user manual
32. APP/UI High-contrast 6-colour key groups (digits, operators, functions, memory, clear, modes) with a "Classic" toggle
33. APP/UI Three bundled offline fonts: Oxanium (display/titles), Barlow Condensed (key faces), JetBrains Mono (key numbers & helpers)
34. APP/UI Show / hide key numbers with one button (remembered between sessions)
35. APP/UI Copy the current display text to the clipboard
36. APP/UI Physical-keyboard shortcuts: digits, + – * / ^ ( ) . Enter Backspace Esc arrows, s c t l n r ! % e p q i
37. APP/UI Fit / 1:1 zoom that never pushes keys off a phone screen
38. APP/UI Reference photo viewer and tool-tip name on every key
39. APP/UI Reset = fresh power-on state; works fully offline (PWA cache / Android APK)

40. PLANNED **Stack history view**
41. PLANNED **Undo**
42. PLANNED **Equation library**
43. PLANNED **Speech read-out**
44. PLANNED **Colour-blind key theme**

2b · Feature guide

Calculator #4 is a RPL: objects go on an unlimited stack with ENTER; commands act on level 1, 2, …; algebraics in quotes. Each card below is one of its 30 working features, with the keys that reach it and, where one exists, a verified example.

F1 Unlimited-depth stack with six visible levels and scrolling ▲ ▼

Where: #50 n! Factorial n! (also on non-integers via the gamma function where supported); #10 ▲ Scroll stack up; #16 ▼ Scroll stack down

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F2 Command line with editing ⌫ before ENTER

Where: #37 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #39 EEX EEX — enter a power-of-ten exponent for the number in X

See it in action: Basic RPL arithmetic — *2 ENTER 3 +* = 5; Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 → LIST Σ LIST* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F3 Six soft-menu keys F1–F6 with six menu pages (NXT / PREV)

Where: #9 NXT Next soft-menu page; #1 F1 Soft key 1 (label on screen); #2 F2 Soft key 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F4 DUP, DROP, SWAP, OVER, ROT, UNROT

Where: #21 SWAP Swap levels 1 and 2; #19 DUP Duplicate level 1; #20 DROP Drop level 1

See it in action: DUP then multiply (square) — *7 DUP ×* = 49; OVER copies level 2 — *2 ENTER 5 OVER +* = 7

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F5 PICK, ROLL, ROLLD, DEPTH

Where: #22 OVER Copy level 2; #23 ROT Level 3 to level 1; #50 n! Factorial n! (also on non-integers via the gamma function where supported)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F6 DUP2, DROP2, NIP and CLEAR

Where: #19 DUP Duplicate level 1; #20 DROP Drop level 1; #21 SWAP Swap levels 1 and 2

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F7 Lists { … } entered on the command line

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F8 →LIST and LIST→ build and explode lists

Where: #24 →LIST n items into a list; #14 { } Start a list; #25 ΣLIST Sum of list

See it in action: Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST Σ LIST* = 6; Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F9 ΣLIST and ΠLIST

Where: #25 ΣLIST Sum of list

See it in action: Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST Σ LIST* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F10 SORT and REVLIST

Where: #26 SORT Sort a list

See it in action: Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F11 SIZE, GET, HEAD, TAIL

Where: #27 HEAD First element

See it in action: Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F12 ΔLIST first differences and SEQ 1...n

Where: #14 { } Start a list; #24 →LIST n items into a list; #25 ΣLIST Sum of list

See it in action: Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST ΣLIST* = 6; Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F13 Element-wise list arithmetic with + − × ÷ and scalars

Where: #14 { } Start a list; #24 →LIST n items into a list; #25 ΣLIST Sum of list

See it in action: Basic RPL arithmetic — *2 ENTER 3 +* = 5; Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST ΣLIST* = 6

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F14 Vectors […] with DOT, CROSS and NORM

Where: #60 CROSS Vector cross product; #57 . Vector dot product; #58 DOT Vector dot product

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F15 MEAN, MAX and MIN of a list

Where: #14 { } Start a list; #24 →LIST n items into a list; #25 ΣLIST Sum of list

See it in action: Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST ΣLIST* = 6; Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F16 Symbolic algebraics in quotes, e.g. '2+3*4'

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F17 Arithmetic on algebraics builds a new algebraic

Where: #13 '' Start an algebraic

See it in action: Basic RPL arithmetic — *2 ENTER 3 +* = 5; Evaluate an algebraic — *'2+3*4' EVAL* = 14

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F18 EVAL / →NUM evaluates an algebraic with stored variables

Where: #17 EVAL Evaluate level 1; #13 '' Start an algebraic; #36 RUN Run recorded program

See it in action: Evaluate an algebraic — *'2+3*4' EVAL* = 14

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F19 Named variables: value, 'NAME', STO; RCL and typing the name recalls it

Where: #18 STO STO — store the value into a memory / variable (press the target next); #36 RUN Run recorded program

See it in action: Named variable — *5 'A' STO A 2 ×* = 10

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F20 Registers R0–R9 via STO/RCL + digit

Where: #18 STO STO — store the value into a memory / variable (press the target next); #37 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value; #40 ⌫ Backspace — delete the last digit typed

See it in action: Named variable — *5 'A' STO A 2 ×* = 10

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F21 VARS list and PURGE (one or all)

Where: #36 RUN Run recorded program; #14 { } Start a list; #24 →LIST n items into a list

See it in action: Build a list and sum it — *1 ENTER 2 ENTER 3 ENTER 3 →LIST Σ LIST* = 6; Sort a list, take its head — *{3 1 2} SORT HEAD* = 1

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F22 Keystroke program recorder REC with RUN and step VIEW

Where: #35 REC Record keystrokes; #36 RUN Run recorded program

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F23 UNDO of the last stack operation

Where: #11 UNDO undo; #40 ⌫ Backspace — delete the last digit typed; #10 ▲ Scroll stack up

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F24 LASTARG (LAST X)

Where: #11 UNDO undo; #40 ⌫ Backspace — delete the last digit typed

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F25 Alpha keyboard for names (ALPHA layer)

Where: #8 ALPHA ALPHA — the next key types its letter/variable (red/green label)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F26 Scientific functions sin cos tan, ln, log, e^x, 10^x, $\sqrt{}$, x^2, 1/x, n!

Where: #30 SIN Sine (uses the current angle unit); #31 COS Cosine (current angle unit); #32 TAN Tangent (current angle unit)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F27 Degree / radian / grad modes

Where: #45 DRG DRG — choose/convert the angle unit (Deg / Rad / Grad)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F28 Clear error texts: Too Few Arguments, Bad Argument Type

Where: #11 UNDO undo; #40 ⌫ Backspace — delete the last digit typed

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F29 Numbers in standard, fixed, scientific and engineering formats

Where: from the TOOLS / menu screens of this model (see chapter 4 for the menu key)

Try it: Press Reset, use the key(s) above, and write the display here:

_____

Your notes:

_____

_____

F30 Matrix entry [[a b][c d]] on the command line

Where: #37 ENTER ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

Try it: Press Reset, use the key(s) above, and write the display here:

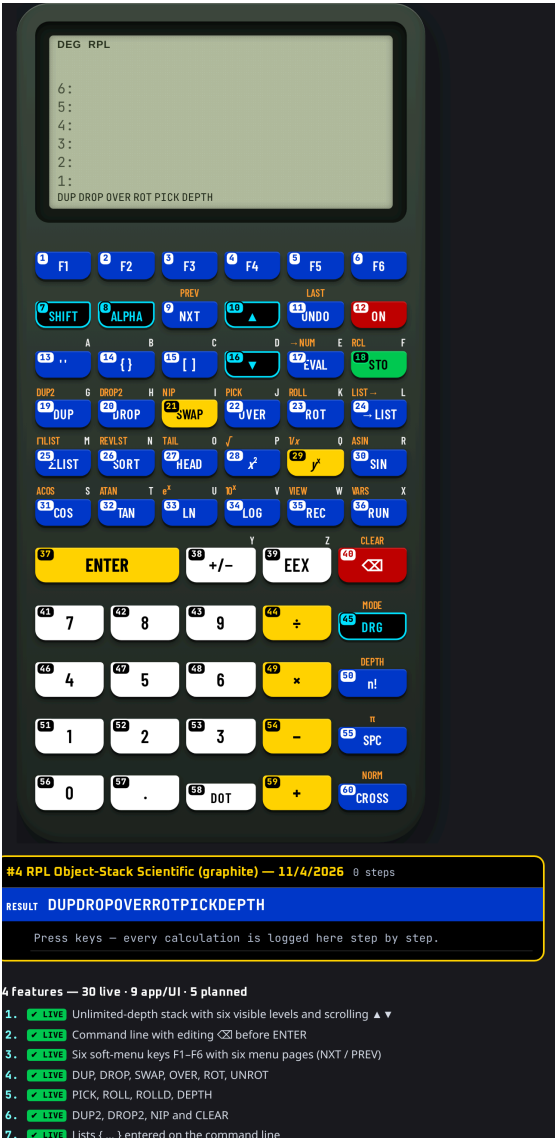
_____

Your notes:

_____

_____

3 · Numbered key map



Calculator #4 with key numbers shown (Hi-Contrast colours).

Key table

“Shift” = 2nd / f / SHIFT layer; “Alpha” = ALPHA layer; “3rd” = 3rd / g layer. A dot means the layer is empty on that key.

#	Face	Shift	Alpha	3rd	Group
1	F1	·	·	·	fn
2	F2	·	·	·	fn
3	F3	·	·	·	fn
4	F4	·	·	·	fn
5	F5	·	·	·	fn
6	F6	·	·	·	fn
7	SHIFT	·	·	·	mod
8	ALPHA	·	·	·	mod
9	NXT	PREV	·	·	fn
10	▲	·	·	·	mod
11	UNDO	LAST	·	·	fn
12	ON	·	·	·	clr
13	''	·	A	·	fn
14	{ }	·	B	·	fn
15	[ ]	·	C	·	fn
16	▼	·	D	·	mod
17	EVAL	→NUM	E	·	fn
18	STO	RCL	F	·	mem
19	DUP	DUP2	G	·	fn
20	DROP	DROP2	H	·	fn
21	SWAP	NIP	I	·	op
22	OVER	PICK	J	·	fn
23	ROT	ROLL	K	·	fn

#	Face	Shift	Alpha	3rd	Group
24	→LIST	LIST→	L	·	fn
25	ΣLIST	ΠLIST	M	·	fn
26	SORT	REVLST	N	·	fn
27	HEAD	TAIL	O	·	fn
28	x^2	√	P	·	fn
29	yx	1/x	Q	·	op
30	SIN	ASIN	R	·	fn
31	COS	ACOS	S	·	fn
32	TAN	ATAN	T	·	fn
33	LN	ex	U	·	fn
34	LOG	10x	V	·	fn
35	REC	VIEW	W	·	fn
36	RUN	VARS	X	·	fn
37	ENTER	·	·	·	op
38	+/−	·	Y	·	num
39	EEX	·	Z	·	num
40	⌫	CLEAR	·	·	clr
41	7	·	·	·	num
42	8	·	·	·	num
43	9	·	·	·	num
44	÷	·	·	·	op
45	DRG	MODE	·	·	mod
46	4	·	·	·	num
47	5	·	·	·	num
48	6	·	·	·	num
49	×	·	·	·	op
50	n!	DEPTH	·	·	fn
51	1	·	·	·	num

#	Face	Shift	Alpha	3rd	Group
52	2	·	·	·	num
53	3	·	·	·	num
54	−	·	·	·	op
55	SPC	π	·	·	fn
56	0	·	·	·	num
57	.	·	·	·	num
58	DOT	·	·	·	num
59	+	·	·	·	op
60	CROSS	NORM	·	·	fn

4 · Key-by-key reference

Every key on calculator #4, in number order, with every function it carries in this app.

#1 F1 Functions

Primary: Soft key 1 (label on screen)

#2 F2 Functions

Primary: Soft key 2

#3 F3 Functions

Primary: Soft key 3

#4 F4 Functions

Primary: Soft key 4

#5 F5 Functions

Primary: Soft key 5

#6 F6 Functions

Primary: Soft key 6

#7 SHIFT `Shift, modes, menus & navigation`

Primary: SHIFT / 2nd — the next key uses its upper (shift-coloured) function

#8 ALPHA `Shift, modes, menus & navigation`

Primary: ALPHA — the next key types its letter/variable (red/green label)

#9 NXT `Functions`

Primary: Next soft-menu page

shifted (2nd / f / SHIFT layer) (press #7 first): PREV — Previous soft-menu page

#10 ▲ `Shift, modes, menus & navigation`

Primary: Scroll stack up

Keyboard: `ArrowUp`

#11 UNDO `Functions`

Primary: undo

shifted (2nd / f / SHIFT layer) (press #7 first): LAST — Last argument

#12 ON `Clear, delete & power`

Primary: ON — power on

#13 '' Functions

Primary: Start an algebraic

alpha layer (press #8 first): A — chr:A

#14 { } Functions

Primary: Start a list

alpha layer (press #8 first): B — chr:B

#15 [ ] Functions

Primary: Start a vector

alpha layer (press #8 first): C — chr:C

#16 ▼ Shift, modes, menus & navigation

Primary: Scroll stack down

alpha layer (press #8 first): D — chr:D

Keyboard: `ArrowDown`

#17 EVAL Functions

Primary: Evaluate level 1

shifted (2nd / f / SHIFT layer) (press #7 first): → NUM — Numeric value

alpha layer (press #8 first): E — chr:E

#18 STO Memory, variables & registers

Primary: STO — store the value into a memory / variable (press the target next)

shifted (2nd / f / SHIFT layer) (press #7 first): RCL — Recall

alpha layer (press #8 first): F — chr:F

#19 DUP Functions

Primary: Duplicate level 1

shifted (2nd / f / SHIFT layer) (press #7 first): DUP2 — Duplicate levels 1–2

alpha layer (press #8 first): G — chr:G

#20 DROP Functions

Primary: Drop level 1

shifted (2nd / f / SHIFT layer) (press #7 first): DROP2 — Drop levels 1–2

alpha layer (press #8 first): H — chr:H

#21 SWAP Operators & execute

Primary: Swap levels 1 and 2

shifted (2nd / f / SHIFT layer) (press #7 first): NIP — Drop level 2

alpha layer (press #8 first): I — chr:I

#22 OVER Functions

Primary: Copy level 2

shifted (2nd / f / SHIFT layer) (press #7 first): PICK — Copy level n

alpha layer (press #8 first): J — chr:J

#23 ROT Functions

Primary: Level 3 to level 1

shifted (2nd / f / SHIFT layer) (press #7 first): ROLL — Roll level n to 1

alpha layer (press #8 first): K — chr:K

#24 →LIST Functions

Primary: n items into a list

shifted (2nd / f / SHIFT layer) (press #7 first): LIST→ — Explode a list

alpha layer (press #8 first): L — chr:L

#25 ΣLIST Functions

Primary: Sum of list

shifted (2nd / f / SHIFT layer) (press #7 first): ΠLIST — Product of list

alpha layer (press #8 first): M — chr:M

#26 SORT Functions

Primary: Sort a list

shifted (2nd / f / SHIFT layer) (press #7 first): REVLST — Reverse a list

alpha layer (press #8 first): N — chr:N

#27 HEAD Functions

Primary: First element

shifted (2nd / f / SHIFT layer) (press #7 first): TAIL — All but first

alpha layer (press #8 first): O — chr:O

#28 x^2 Functions

Primary: Square x^2

shifted (2nd / f / SHIFT layer) (press #7 first): √ — Square root √

alpha layer (press #8 first): P — chr:P

Keyboard: q

#29 yx Operators & execute

Primary: Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)

shifted (2nd / f / SHIFT layer) (press #7 first): 1/x — Reciprocal x^{-1} (1/x)

alpha layer (press #8 first): Q — chr:Q

Keyboard: ^

#30 SIN Functions

Primary: Sine (uses the current angle unit)

shifted (2nd / f / SHIFT layer) (press #7 first): ASIN — Inverse sine $\sin^{-1}$

alpha layer (press #8 first): R — chr:R

Keyboard: s

#31 COS Functions

Primary: Cosine (current angle unit)

shifted (2nd / f / SHIFT layer) (press #7 first): ACOS — Inverse cosine $\cos^{-1}$

alpha layer (press #8 first): S — chr:S

Keyboard: c

#32 TAN Functions

Primary: Tangent (current angle unit)

shifted (2nd / f / SHIFT layer) (press #7 first): ATAN — Inverse tangent $\tan^{-1}$

alpha layer (press #8 first): T — chr:T

Keyboard: t

#33 LN Functions

Primary: Natural logarithm ln

shifted (2nd / f / SHIFT layer) (press #7 first): ex — e^x — exponential

alpha layer (press #8 first): U — chr:U

Keyboard: n

#34 LOG Functions

Primary: Common logarithm $\log_{10}$

shifted (2nd / f / SHIFT layer) (press #7 first): 10x — 10^x — antilog

alpha layer (press #8 first): V — chr:V

Keyboard: `l`

#35 REC Functions

Primary: Record keystrokes

shifted (2nd / f / SHIFT layer) (press #7 first): VIEW — View program

alpha layer (press #8 first): W — chr:W

#36 RUN Functions

Primary: Run recorded program

shifted (2nd / f / SHIFT layer) (press #7 first): VARS — List variables

alpha layer (press #8 first): X — chr:X

#37 ENTER Operators & execute

Primary: ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value

Keyboard: `Enter`

#38 +/− Digits & number entry

Primary: Change sign (+/−) of the number being typed or the current result

alpha layer (press #8 first): Y — chr:Y

#39 EEX Digits & number entry

Primary: EEX — enter a power-of-ten exponent for the number in X

alpha layer (press #8 first): Z — chr:Z

Keyboard: e

#40 ⌫ Clear, delete & power

Primary: Backspace — delete the last digit typed

shifted (2nd / f / SHIFT layer) (press #7 first): CLEAR — Clear stack

Keyboard: Backspace Delete

#41 7 Digits & number entry

Primary: Digit 7

Keyboard: 7

#42 8 Digits & number entry

Primary: Digit 8

Keyboard: 8

#43 **9** `Digits & number entry`

Primary: Digit 9

Keyboard: 9

#44 **÷** `Operators & execute`

Primary: Division ÷

Keyboard: /

#45 **DRG** `Shift, modes, menus & navigation`

Primary: DRG — choose/convert the angle unit (Deg / Rad / Grad)

shifted (2nd / f / SHIFT layer) (press #7 first): MODE — Display format

#46 **4** `Digits & number entry`

Primary: Digit 4

Keyboard: 4

#47 **5** `Digits & number entry`

Primary: Digit 5

Keyboard: 5

#48 **6** `Digits & number entry`

Primary: Digit 6

Keyboard: 6

#49 × Operators & execute

Primary: Multiplication ×

Keyboard: * x

#50 n! Functions

Primary: Factorial n! (also on non-integers via the gamma function where supported)

shifted (2nd / f / SHIFT layer) (press #7 first): DEPTH — Stack depth

Keyboard: !

#51 1 Digits & number entry

Primary: Digit 1

Keyboard: 1

#52 2 Digits & number entry

Primary: Digit 2

Keyboard: 2

#53 3 Digits & number entry

Primary: Digit 3

Keyboard: 3

#54 – Operators & execute

Primary: Subtraction –

Keyboard: -

#55 SPC Functions

Primary: Space separator

shifted (2nd / f / SHIFT layer) (press #7 first): π — π (3.14159265…) as an exact constant

#56 0 Digits & number entry

Primary: Digit 0

Keyboard: 0

#57 . Digits & number entry

Primary: Vector dot product

Keyboard: . ,

#58 DOT Digits & number entry

Primary: Vector dot product

#59 + Operators & execute

Primary: Addition +

Keyboard: +

#60 CROSS Functions

Primary: Vector cross product

shifted (2nd / f / SHIFT layer) (press #7 first): NORM — Vector length

5 · Worked examples

8 examples, each run on this calculator's keys by the U-45 checker. Follow the chips in order.

Example 5.1 — Basic RPL arithmetic

Calculate **2 ENTER 3 +**. Press **Reset** first.

#52 2 → #37 ENTER → #53 3 → #59 +

Display shows `1:5` — expected 5. ✔ verified on the app.

Example 5.2 — DUP then multiply (square)

Calculate **7 DUP ×**. Press **Reset** first.

#41 7 → #19 DUP → #49 ×

Display shows `1:49` — expected 49. ✔ verified on the app.

Example 5.3 — Build a list and sum it

Calculate **1 ENTER 2 ENTER 3 ENTER 3 →LIST ΣLIST**. Press **Reset** first.

#51 1 → #37 ENTER → #52 2 → #37 ENTER → #53 3 → #37 ENTER → #53 3 → #24 →LIST → #25 ΣLIST

Display shows `1:6` — expected 6. ✔ verified on the app.

Example 5.4 — Evaluate an algebraic

Calculate '2+3*4' EVAL. Press Reset first.

#13 '' → #52 2 → #59 + → #53 3 → #49 × → #46 4 → #37 ENTER → #17 EVAL

Display shows `1:14` — expected 14. ✔ verified on the app.

Example 5.5 — OVER copies level 2

Calculate 2 ENTER 5 OVER +. Press Reset first.

#52 2 → #37 ENTER → #47 5 → #22 OVER → #59 +

Display shows `1:7` — expected 7. ✔ verified on the app.

Example 5.6 — ROT brings level 3 to level 1

Calculate 1 ENTER 2 ENTER 3 ROT. Press Reset first.

#51 1 → #37 ENTER → #52 2 → #37 ENTER → #53 3 → #23 ROT

Display shows `1:1` — expected 1. ✔ verified on the app.

Example 5.7 — Sort a list, take its head

Calculate **{3 1 2} SORT HEAD**. Press **Reset** first.

#14 { } → #53 3 → #55 SPC → #51 1 → #55 SPC → #52 2 → #37 ENTER → #26 SORT → #27 HEAD

Display shows `1:1` — expected 1. ✔ verified on the app.

Example 5.8 — Named variable

Calculate **5 'A' STO A 2 ×**. Press **Reset** first.

#47 5 → #37 ENTER → #13 '' → #8 ALPHA → #13 A → #37 ENTER → #18 STO → #8 ALPHA → #13 A → #37 ENTER → #52 2 → #49 ×

Display shows `1:10` — expected 10. ✔ verified on the app.

6 · Tricks & hacks

Hack 1: No equals key

RPN works number, number, operator: 3 ENTER 4 + = 7. Results stay on the stack for the next step.

Hack 2: Chain without ENTER

After an operation, just type the next number and operator: 3 ENTER 4 + 5 × gives (3+4)×5 = 35.

Hack 3: Swap to fix order

If you entered numbers in the wrong order for − ÷ or y^x, press x≷y before the operator.

Hack 4: LASTx to reuse

LASTx brings back the X value of the previous operation — useful for a × a.

Hack 5: Stack as memory

Up to four values are kept; R↓ rolls through them.

Hack 6: Constant via stack top

On 4-level stacks the T register copies down, so a constant multiplier can be reused by pressing × repeatedly.

Hack 7: Hidden functions on this model

53 extra functions sit on shifted layers. Press the layer key once, then the key: for example #7 then #9 → PREV; #7 then #11 → LAST; #8 then #13 → A; #8 then #14 → B.

Hack 8: Find any key by number

Use Appendix B (function finder): look up the function, read its key number, then find that number on the keypad. Numbers run left → right, top → bottom from #1 to #60.

Hack 9: Type faster on a computer

28 keys on this model have a keyboard shortcut (see chapter 7). Typing digits and operators is much faster than clicking.

Hack 10: Check before you trust

Estimate first (round each number) and compare with the display. If the answer is 10× or 100× off, a decimal point or percent key was missed.

Hack 11: Copy results out

Press Copy in the bar above the calculator to put the display text on the clipboard instead of retyping it.

Hack 12: Colour blind or low vision?

Hi-Contrast colours use black/white/yellow/blue/green/red with black or white text for maximum contrast; key numbers stay on in both colour schemes.

7 · Keyboard shortcuts

With the calculator open on a computer, these physical keys press the numbered keys below (Ctrl/Alt/⌘ combinations are left to the browser). Shifted functions are reached by clicking the shift key first.

Keyboard	#	Key	Does
ArrowUp	#10	▲	Scroll stack up
ArrowDown	#16	▼	Scroll stack down
q	#28	x^2	Square x^2
^	#29	yx	Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)
s	#30	SIN	Sine (uses the current angle unit)
c	#31	COS	Cosine (current angle unit)
t	#32	TAN	Tangent (current angle unit)
n	#33	LN	Natural logarithm ln
l	#34	LOG	Common logarithm $\log_{10}$
Enter	#37	ENTER	ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value
e	#39	EEX	EEX — enter a power-of-ten exponent for the number in X
Backspace Delete	#40	⌫	Backspace — delete the last digit typed
7	#41	7	Digit 7
8	#42	8	Digit 8
9	#43	9	Digit 9
/	#44	÷	Division ÷
4	#46	4	Digit 4
5	#47	5	Digit 5
6	#48	6	Digit 6
* x	#49	×	Multiplication ×

!	#50	n!	Factorial n! (also on non-integers via the gamma function where supported)
1	#51	1	Digit 1
2	#52	2	Digit 2
3	#53	3	Digit 3
-	#54	–	Subtraction –
0	#56	0	Digit 0
. ,	#57	.	Vector dot product
+	#59	+	Addition +

App shortcuts

“# On / # Off”	show / hide key numbers
“Classic / Hi-Contrast”	switch key colours
“Copy”	copy display text
“Fit / 1:1”	scale to screen / design size
“Reset”	restart this calculator
Jump list	switch to another of the 45 calculators

8 · Common mistakes

Mistake 1: Pressing ENTER after an operation

ENTER after + duplicates the result. Usually you want to type the next number directly.

Mistake 2: Wrong order for – and ÷

Y – X and Y ÷ X: enter the first operand first. 10 ENTER 4 – = 6.

Mistake 3: CHS vs –

Make a number negative with CHS/+/–, not with –.

Mistake 4: Prefix left active

If f / g / SHIFT is shown, the next key is shifted. Clear it with the prefix-clear function.

Mistake 5: Forgetting Reset between problems

Old values (memory, stack, open brackets, worksheet values) can leak into the next calculation. Press Reset or clear before starting a new problem.

Mistake 6: Pressing the wrong layer

A shift / 2nd indicator that is still lit will turn the next key into its shifted function. Look at the display indicators before pressing.

9 · Practice drills

Set A — Calculation drills (8)

Work each one on calculator #4, then check your keys and the answer in Appendix A. Every answer was produced by this calculator.

1. 2 ENTER 3 + = _____
2. 7 DUP × = _____
3. 1 ENTER 2 ENTER 3 ENTER 3 →LIST ΣLIST = _____
4. '2+3*4' EVAL = _____
5. 2 ENTER 5 OVER + = _____
6. 1 ENTER 2 ENTER 3 ROT = _____
7. {3 1 2} SORT HEAD = _____
8. 5 'A' STO A 2 × = _____

Set B — Key-finding drills (103)

Without looking at the table, find each key on the picture. Answers in Appendix A.

1. Which key is “F1” (Soft key 1)? _____
2. Which key is “F2” (Soft key 2)? _____
3. Which key is “F3” (Soft key 3)? _____
4. Which key is “F4” (Soft key 4)? _____
5. Which key is “F5” (Soft key 5)? _____
6. Which key is “F6” (Soft key 6)? _____

7. Which key is “SHIFT” (SHIFT / 2nd)? _____

8. Which key is “ALPHA” (ALPHA)? _____

9. Which key is “NXT” (Next soft-menu page)? _____

10. Which key is “ ▲ ” (Scroll stack up)? _____

11. Which key is “UNDO” (undo)? _____

12. Which key is “ON” (ON)? _____

13. Which key is “' '” (Start an algebraic)? _____

14. Which key is “{ }” (Start a list)? _____

15. Which key is “[ ]” (Start a vector)? _____

16. Which key is “ ▼ ” (Scroll stack down)? _____

17. Which key is “EVAL” (Evaluate level 1)? _____

18. Which key is “STO” (STO)? _____

19. Which key is “DUP” (Duplicate level 1)? _____

20. Which key is “DROP” (Drop level 1)? _____

21. Which key is “SWAP” (Swap levels 1 and 2)? _____

22. Which key is “OVER” (Copy level 2)? _____

23. Which key is “ROT” (Level 3 to level 1)? _____

24. Which key is “→LIST” (n items into a list)? _____

25. Which key is “ΣLIST” (Sum of list)? _____

26. Which key is “SORT” (Sort a list)? _____

27. Which key is “HEAD” (First element)? _____

28. Which key is “x^2” (Square x^2)? _____

29. Which key is “yx” (Power x^y / ^)? _____

30. Which key is “SIN” (Sine)? _____

31. Which key is “COS” (Cosine)? _____

32. Which key is “TAN” (Tangent)? _____

33. Which key is “LN” (Natural logarithm ln)? _____

34. Which key is “LOG” (Common logarithm $\log_{10}$)? _____

35. Which key is “REC” (Record keystrokes)? _____

36. Which key is “RUN” (Run recorded program)? _____

37. Which key is “ENTER” (ENTER)? _____

38. Which key is “+/–” (Change sign)? _____

39. Which key is “EEX” (EEX)? _____

40. Which key is “⌫” (Backspace)? _____

41. Which key is “÷” (Division ÷)? _____

42. Which key is “DRG” (DRG)? _____

43. Which key is “×” (Multiplication ×)? _____

44. Which key is “n!” (Factorial n!)? _____

45. Which key is “–” (Subtraction –)? _____

46. Which key is “SPC” (Space separator)? _____

47. Which key is "." (Vector dot product)? _____

48. Which key is "DOT" (Vector dot product)? _____

49. Which key is "+" (Addition +)? _____

50. Which key is "CROSS" (Vector cross product)? _____

51. How do you reach "PREV"? _____

52. How do you reach "LAST"? _____

53. How do you reach "A"? _____

54. How do you reach "B"? _____

55. How do you reach "C"? _____

56. How do you reach "D"? _____

57. How do you reach " → NUM"? _____

58. How do you reach "E"? _____

59. How do you reach "RCL"? _____

60. How do you reach "F"? _____

61. How do you reach "DUP2"? _____

62. How do you reach "G"? _____

63. How do you reach "DROP2"? _____

64. How do you reach "H"? _____

65. How do you reach "NIP"? _____

66. How do you reach "I"? _____

67. How do you reach “PICK”? _____

68. How do you reach “J”? _____

69. How do you reach “ROLL”? _____

70. How do you reach “K”? _____

71. How do you reach “LIST →”? _____

72. How do you reach “L”? _____

73. How do you reach “⊓LIST”? _____

74. How do you reach “M”? _____

75. How do you reach “REVLST”? _____

76. How do you reach “N”? _____

77. How do you reach “TAIL”? _____

78. How do you reach “O”? _____

79. How do you reach “√”? _____

80. How do you reach “P”? _____

81. How do you reach “1/x”? _____

82. How do you reach “Q”? _____

83. How do you reach “ASIN”? _____

84. How do you reach “R”? _____

85. How do you reach “ACOS”? _____

86. How do you reach “S”? _____

87. How do you reach “ATAN”? _____

88. How do you reach “T”? _____

89. How do you reach “ex”? _____

90. How do you reach “U”? _____

91. How do you reach “10x”? _____

92. How do you reach “V”? _____

93. How do you reach “VIEW”? _____

94. How do you reach “W”? _____

95. How do you reach “VARS”? _____

96. How do you reach “X”? _____

97. How do you reach “Y”? _____

98. How do you reach “Z”? _____

99. How do you reach “CLEAR”? _____

100. How do you reach “MODE”? _____

101. How do you reach “DEPTH”? _____

102. How do you reach “π”? _____

103. How do you reach “NORM”? _____

Appendix A — Drill answers

Set A

1. **5** (display `1:5`)

   #52 2 → #37 ENTER → #53 3 → #59 +

2. **49** (display `1:49`)

   #41 7 → #19 DUP → #49 ×

3. **6** (display `1:6`)

   #51 1 → #37 ENTER → #52 2 → #37 ENTER → #53 3 → #37 ENTER → #53 3 → #24 →LIST → #25 ΣLIST

4. **14** (display `1:14`)

   #13 '' → #52 2 → #59 + → #53 3 → #49 × → #46 4 → #37 ENTER → #17 EVAL

5. **7** (display `1:7`)

   #52 2 → #37 ENTER → #47 5 → #22 OVER → #59 +

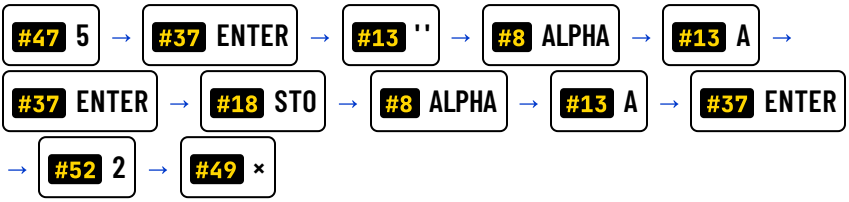
6. **1** (display `1:1`)

   #51 1 → #37 ENTER → #52 2 → #37 ENTER → #53 3 → #23 ROT

7. **1** (display `1:1`)

   #14 { } → #53 3 → #55 SPC → #51 1 → #55 SPC → #52 2 → #37 ENTER → #26 SORT → #27 HEAD

8. **10** (display `1:10`)



Set B

1. #1
2. #2
3. #3
4. #4
5. #5
6. #6
7. #7
8. #8
9. #9
10. #10
11. #11
12. #12
13. #13
14. #14
15. #15
16. #16
17. #17
18. #18
19. #19
20. #20
21. #21
22. #22
23. #23
24. #24
25. #25
26. #26
27. #27
28. #28
29. #29
30. #30
31. #31
32. #32
33. #33
34. #34
35. #35
36. #36
37. #37
38. #38
39. #39
40. #40
41. #44
42. #45
43. #49
44. #50
45. #54
46. #55
47. #57
48. #58
49. #59
50. #60
51. #7 SHIFT then #9 NXT
52. #7 SHIFT then #11 UNDO
53. #8 ALPHA then #13 ' '
54. #8 ALPHA then #14 { }
55. #8 ALPHA then #15 [ ]
56. #8 ALPHA then #16 ▼
57. #7 SHIFT then #17 EVAL
58. #8 ALPHA then #17 EVAL
59. #7 SHIFT then #18 STO
60. #8 ALPHA then #18 STO
61. #7 SHIFT then #19 DUP
62. #8 ALPHA then #19 DUP

63. #7 SHIFT then #20 DROP
64. #8 ALPHA then #20 DROP
65. #7 SHIFT then #21 SWAP
66. #8 ALPHA then #21 SWAP
67. #7 SHIFT then #22 OVER
68. #8 ALPHA then #22 OVER
69. #7 SHIFT then #23 ROT
70. #8 ALPHA then #23 ROT
71. #7 SHIFT then #24 →LIST
72. #8 ALPHA then #24 →LIST
73. #7 SHIFT then #25 ΣLIST
74. #8 ALPHA then #25 ΣLIST
75. #7 SHIFT then #26 SORT
76. #8 ALPHA then #26 SORT
77. #7 SHIFT then #27 HEAD
78. #8 ALPHA then #27 HEAD
79. #7 SHIFT then #28 x²
80. #8 ALPHA then #28 x²
81. #7 SHIFT then #29 yx
82. #8 ALPHA then #29 yx
83. #7 SHIFT then #30 SIN
84. #8 ALPHA then #30 SIN
85. #7 SHIFT then #31 COS
86. #8 ALPHA then #31 COS
87. #7 SHIFT then #32 TAN
88. #8 ALPHA then #32 TAN
89. #7 SHIFT then #33 LN
90. #8 ALPHA then #33 LN
91. #7 SHIFT then #34 LOG
92. #8 ALPHA then #34 LOG
93. #7 SHIFT then #35 REC
94. #8 ALPHA then #35 REC
95. #7 SHIFT then #36 RUN
96. #8 ALPHA then #36 RUN
97. #8 ALPHA then #38 +/-
98. #8 ALPHA then #39 EEX
99. #7 SHIFT then #40 ⌫
100. #7 SHIFT then #45 DRG
101. #7 SHIFT then #50 n!
102. #7 SHIFT then #55 SPC
103. #7 SHIFT then #60 CROSS

Appendix B — Function finder (A–Z)

Function	Label	Keys
10^x — antilog	10x	#7 → #34
Addition +	+	#59
All but first	TAIL	#7 → #27
ALPHA — the next key types its letter/variable (red/green label)	ALPHA	#8
Backspace — delete the last digit typed	⌫	#40
Change sign (+/−) of the number being typed or the current result	+/−	#38
chr:A	A	#8 → #13
chr:B	B	#8 → #14
chr:C	C	#8 → #15
chr:D	D	#8 → #16
chr:E	E	#8 → #17
chr:F	F	#8 → #18
chr:G	G	#8 → #19
chr:H	H	#8 → #20
chr:I	I	#8 → #21
chr:J	J	#8 → #22

chr:K	**K**	**#8 → #23**
chr:L	**L**	**#8 → #24**
chr:M	**M**	**#8 → #25**
chr:N	**N**	**#8 → #26**
chr:O	**O**	**#8 → #27**
chr:P	**P**	**#8 → #28**
chr:Q	**Q**	**#8 → #29**
chr:R	**R**	**#8 → #30**
chr:S	**S**	**#8 → #31**
chr:T	**T**	**#8 → #32**
chr:U	**U**	**#8 → #33**
chr:V	**V**	**#8 → #34**
chr:W	**W**	**#8 → #35**
chr:X	**X**	**#8 → #36**
chr:Y	**Y**	**#8 → #38**
chr:Z	**Z**	**#8 → #39**

Clear stack	CLEAR	#7 → #40
Common logarithm $\log_{10}$	LOG	#34
Copy level 2	OVER	#22
Copy level n	PICK	#7 → #22
Cosine (current angle unit)	COS	#31
Digit 0	0	#56
Digit 1	1	#51
Digit 2	2	#52
Digit 3	3	#53
Digit 4	4	#46
Digit 5	5	#47
Digit 6	6	#48
Digit 7	7	#41
Digit 8	8	#42
Digit 9	9	#43
Display format	MODE	#7 → #45
Division ÷	÷	#44
DRG — choose/convert the angle unit (Deg / Rad / Grad)	DRG	#45
Drop level 1	DROP	#20
Drop level 2	NIP	#7 → #21
Drop levels 1–2	DROP2	#7 → #20
Duplicate level 1	DUP	#19
Duplicate levels 1–2	DUP2	#7 → #19
EEX — enter a power-of-ten exponent for the number in X	EEX	#39

ENTER — terminates digit entry and pushes X up the stack (RPN) / confirms a value	ENTER	#37
Evaluate level 1	EVAL	#17
e^x — exponential	ex	#7 → #33
Explode a list	LIST→	#7 → #24
Factorial n! (also on non-integers via the gamma function where supported)	n!	#50
First element	HEAD	#27
Inverse cosine $\cos^{-1}$	ACOS	#7 → #31
Inverse sine $\sin^{-1}$	ASIN	#7 → #30
Inverse tangent $\tan^{-1}$	ATAN	#7 → #32
Last argument	LAST	#7 → #11
Level 3 to level 1	ROT	#23
List variables	VARS	#7 → #36
Multiplication ×	×	#49
n items into a list	→LIST	#24
Natural logarithm ln	LN	#33
Next soft-menu page	NXT	#9
Numeric value	→NUM	#7 → #17
ON — power on	ON	#12
Power x^y / ^ — raise to any power. Chains right-to-left (2^3^2 = 2^9 = 512)	yx	#29
Previous soft-menu page	PREV	#7 → #9
Product of list	ΠLIST	#7 → #25

Recall	RCL	#7 → #18
Reciprocal x^{-1} (1/x)	1/x	#7 → #29
Record keystrokes	REC	#35
Reverse a list	REVLST	#7 → #26
Roll level n to 1	ROLL	#7 → #23
Run recorded program	RUN	#36
Scroll stack down	▼	#16
Scroll stack up	▲	#10
SHIFT / 2nd — the next key uses its upper (shift-coloured) function	SHIFT	#7
Sine (uses the current angle unit)	SIN	#30
Soft key 1 (label on screen)	F1	#1
Soft key 2	F2	#2
Soft key 3	F3	#3
Soft key 4	F4	#4
Soft key 5	F5	#5
Soft key 6	F6	#6
Sort a list	SORT	#26
Space separator	SPC	#55
Square root √	√	#7 → #28
Square x^2	x^2	#28
Stack depth	DEPTH	#7 → #50
Start a list	{ }	#14
Start a vector	[ ]	#15
Start an algebraic	' '	#13

STO — store the value into a memory / variable (press the target next)	STO	#18
Subtraction −	−	#54
Sum of list	ΣLIST	#25
Swap levels 1 and 2	SWAP	#21
Tangent (current angle unit)	TAN	#32
undo	UNDO	#11
Vector cross product	CROSS	#60
Vector dot product	.	#57
Vector dot product	DOT	#58
Vector length	NORM	#7 → #60
View program	VIEW	#7 → #35
π (3.14159265…) as an exact constant	π	#7 → #55

Appendix C — Glossary

Ans
The result of the last calculation, reusable in the next.

AOS
Algebraic Operating System — operator precedence on a one-line display.

RPN
Reverse Polish Notation — operands first, then the operation; no = key.

Stack
The RPN registers X, Y, Z, T where numbers wait for operations.

TVM
Time Value of Money — n, i, PV, PMT, FV.

Natural display
Shows fractions, roots and powers as printed in textbooks.

Shift / 2nd
Prefix key that selects the function printed above a key.

Alpha
Prefix key that types the letter printed beside a key.

Radian
Angle unit; 2π rad = 360°.

Grad
Angle unit; 400 grad = 360°.

Engineering notation
Scientific notation with exponents that are multiples of 3.

Mantissa
The digits of a number in scientific notation.

Memory (M)

An independent register for running totals.

Mark-up

Increase over cost; margin is profit as a percent of selling price.

Grand total (GT)

Sum of all totals since the last clear.

nCr / nPr

Combinations / permutations of n items taken r at a time.

Factorial

n! = 1×2×…×n.

Reciprocal

1/x.

Standard deviation

Spread of data around the mean (σ population, s sample).

Key number

The small badge 1…N on each key in U-45 CALCULATORS, used throughout this manual.

Appendix D — Model notes & limits

Reverse-Polish-Lisp style stack processor: every object (number, list, vector, algebraic) lives on an unlimited stack. All stack and list logic is original U-45 code.

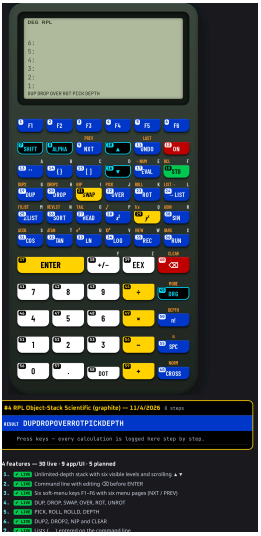
An algebraic typed in quotes stays symbolic until EVAL; arithmetic between algebraics joins them into a new algebraic.

Planned (not yet working)

- Stack history view
- Undo
- Equation library
- Speech read-out
- Colour-blind key theme

U-45 CALCULATORS models are independent software recreations; they are not affiliated with any calculator manufacturer and carry no third-party trademarks.

Appendix E — Quick-reference card



#1 F1	#2 F2	#3 F3
#4 F4	#5 F5	#6 F6
#7 SHIFT	#8 ALPHA	#9 NXT ⇧ PREV
#10 ▲	#11 UNDO ⇧ LAST	#12 ON
#13 ' '	#14 { }	#15 [ ]
#16 ▼	#17 EVAL ⇧ →NUM	#18 STO ⇧ RCL
#19 DUP ⇧ DUP2	#20 DROP ⇧ DROP2	#21 SWAP ⇧ NIP
#22 OVER ⇧ PICK	#23 ROT ⇧ ROLL	#24 →LIST ⇧ LIST→
#25 ΣLIST ⇧ ΠLIST	#26 SORT ⇧ REVLST	#27 HEAD ⇧ TAIL
#28 x^2 ⇧ √	#29 yx ⇧ 1/x	#30 SIN ⇧ ASIN
#31 COS ⇧ ACOS	#32 TAN ⇧ ATAN	#33 LN ⇧ ex
#34 LOG ⇧ 10x	#35 REC ⇧ VIEW	#36 RUN ⇧ VARS
#37 ENTER	#38 +/−	#39 EEX
#40 ⌫ ⇧ CLEAR	#41 7	#42 8
#43 9	#44 ÷	#45 DRG ⇧ MODE
#46 4	#47 5	#48 6
#49 ×	#50 n! ⇧ DEPTH	#51 1

#52 2	#53 3	#54 −
#55 SPC ⇧ π	#56 0	#57 .
#58 DOT	#59 +	#60 CROSS ⇧ NORM

U-45 CALCULATORS · Calculator #4 · cut out and keep